

Problem 1.

a) $v_2 = \frac{m}{m+M} v_1$

b) $h = \left(\frac{m}{m+M} \right)^2 \frac{v_1^2}{2g}$

Problem 2.

D.

Problem 3.

A, E.

Problem 4.

D.

Problem 5.

a) $v = -\frac{3}{5}u$

b) B.

Problem 6.

$$v_1 = \frac{M_2 + m}{M_1 + M_2 + m} v$$

$$v_2 = \frac{m}{M_1 + M_2 + m} v$$

Problem 7.

$$v = 15.6 \frac{\text{m}}{\text{s}}$$

$$\theta = 53.1^\circ$$

Problem 8.

A.

Problem 9.

D.